

Systems biology

Predict drug permeability to blood–brain–barrier from clinical phenotypes: drug side effects and drug indications

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Abstract

Motivation: Blood–Brain–Barrier (BBB) is a rigorous permeability barrier for maintaining homeostasis of Central Nervous System (CNS). Determination of compound's permeability to BBB is prerequisite in CNS drug discovery. Existing computational methods usually predict drug BBB permeability from chemical structure and they generally apply to small compounds passing BBB through passive diffusion. As abundant information on drug side effects and indications has been recorded over time through extensive clinical usage, we aim to explore BBB permeability prediction from a new angle and introduce a novel approach to predict BBB permeability from drug clinical phenotypes (drug side effects and drug indications). This method can apply to both small compounds and macro-molecules penetrating BBB through various mechanisms besides passive diffusion.

Results: We composed a training dataset of 213 drugs with known brain and blood steady-state concentrations ratio and extracted their side effects and indications as features. Next, we trained SVM models with polynomial kernel and obtained accuracy of 76.0%, AUC 0.739, and F₁ score (macro weighted) 0.760 with Monte Carlo cross validation. The independent test accuracy was 68.3%, AUC 0.692, F₁ score 0.676. When both chemical features and clinical phenotypes were available, combining the two types of features achieved significantly better performance than chemical feature based approach (accuracy 85.5% versus 72.9%, AUC 0.854 versus 0.733, F₁ score 0.854 versus 0.725; $P < e^{-90}$). We also conducted *de novo* prediction and identified 110 drugs in SIDER database having the potential to penetrate BBB, which could serve as start point for CNS drug repositioning research.

Availability and Implementation: <https://github.com/bioinformatics-gao/CASE-BBB-prediction-Data>

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Supplementary information: [Supplementary data](#) are available at *Bioinformatics* online.

1 Introduction

Neurologic and psychiatric disorders account for a remarkable 28% of all years of life lived with a disability (Menken *et al.*, 2000). Despite the high prevalence, enormous severity and heavy social-economic burden associated with Central Nervous System (CNS) disorders, effective medicines for these diseases are in scarcity, e.g.

there is essentially no approved drug for the devastating and lethal transmissible spongiform encephalopathies (TSEs) in human and animals (Cordeiro, 2016).

We previously conducted research to explore the possibility of drug repositioning to Parkinson's disease, schizophrenia and glioblastoma (Chen and Xu, 2016; Chen *et al.*, 2015, 2016; Xu and

Wang, 2015). However, many tested compounds failed due to lack of the ability to penetrate Blood-Brain-Barrier (BBB) rather than lack of potency, which made BBB bottleneck in CNS drug discovery (van Tellingen et al., 2015; Weidle, et al., 2015).

BBB in human is formed by specialized tight junctions between endothelial cells that line brain capillaries (Davson, 1989; Hendricks et al., 2015) and it is a rigorous and highly selective permeability barrier separating brain from circulating blood (Daneman and Prat, 2015; Rubin and Staddon, 1999; Saunders et al., 2014). BBB prevents most external compounds (98%) from entering CNS so as to maintain CNS homeostasis (Pardridge, 2005). Therefore, determination of compounds' BBB permeability is a prerequisite for screening compounds/bio-molecules which could take effects in CNS (Daneman and Prat, 2015).

Drug BBB permeability is commonly indicated by the ratio or log-ratio of the drug brain concentration to the blood concentration at steady state (Bickel, 2005; Carpenter et al., 2014). Although *in vivo* experiment is the most accurate approach to measure BBB permeability, it becomes resource-extensive and impossible for large scale drug screening (Bickel, 2005; Saunders et al., 2014). Alternatively, the *in silico* prediction of drug BBB permeability could avoid the tremendous cost and time associated with the *in vivo* experiments and attracted more and more attention (reviewed in Bickel, 2005; Kortagere et al., 2008).

The previous methods predict molecular BBB permeability from physical and chemical features of compounds, including topological polar surface area, van der Waals volume, water-accessible volume, rotatable-bonds, hydrogen-bond donator and acceptor, acidic and basic atoms numbers, ionization potential etc. (Carpenter et al., 2014; Kumar et al., 2013; Li et al., 2005; Norinder and Haeberlein, 2002). Various supervised learning approaches such as linear regression (Liu et al., 2004), multivariate genetic partial least squares (Subramaniana and Kitchen, 2003), neural network (Doniger et al., 2002), support vector machine (Shen et al., 2010) were adopted to train prediction models with physical or chemical features. In general, those methods only apply to the small compounds passing BBB through passive diffusion (Fig. 1, left part) (Kumar et al., 2013; Norinder and Haeberlein, 2002). However, many molecules, e.g. glucose and insulin, pass BBB through more complex mechanism than passive diffusion (Fig. 1, right part) (Banks, 2004; Crone, 1965). Those complex mechanisms for drug passing BBB involve sophisticated molecular recognition for specific drug-transporter/ drug-receptor interaction and are hard to be described by general physical- or chemical-features of compounds. Also, some

chemotherapy drugs/toxins were pumped out from CNS (Fig. 1, right part) although they could infiltrate BBB initially (Fellner et al., 2002).

To address the multiple mechanisms associated with drugs passing BBB and sustaining in the central nervous system, we need to develop new methods or incorporate new features. Up to now, thousands of drugs have been clinically tested and left abundant information on drug clinical phenotypes (drug side effects and drug indications) (Kuhn et al., 2015), which provide us valuable sources for drug BBB prediction. In order to overcome the limitations of chemical features based methods, we proposed the novel approach of predicting drug BBB permeability from drug clinical phenotypes. Compared with chemical structure based method, our drug clinical phenotypes based prediction method has the following advantages: (i) Applicable to small molecular drugs that pass BBB with mechanism other than simple diffusion; (ii) Applicable to macro-molecular drugs which can hardly be handled by current chemical structure based methods; (iii) Applicable to molecules that were actively pumped out from CNS (Fig. 1 right part). Especially, when both drug clinical phenotypes and structure parameters are available, our study demonstrated prediction through combination of the two groups of features produced significantly higher accuracy than using structure features alone (85% versus 73%, $P < e^{-90}$).

Overall, our work demonstrated BBB permeability prediction from drug side effects and indications is a feasible and practical method with high performance: (i) When the drug chemical structure parameters are available, integration of drug clinical phenotypes achieved significantly higher performance than using structure features alone. (ii) When the drug chemical structure parameters are not available, our method is still applicable so as to extend the prediction scope.

2 Methods

We curated drug datasets with known BBB permeability. The side effects and indications of these drugs were extracted from SIDER database. A classification model with SVM algorithm was built and tested. Next we evaluated the performance of the model in independent dataset. Afterwards, we combined drug clinical phenotype and structure features to evaluate if the performance significantly improved. Finally, we conducted *de novo* prediction for drugs in the SIDER database. The schema for the experiment steps was shown in Figure 2.

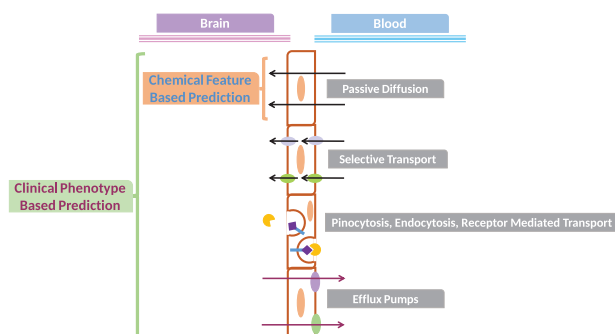


Fig. 1. Mechanisms of drugs passing BBB and the scope of prediction methods. *Right part* shows the mechanisms for drug passing BBB, *left part* shows the scope of drug clinical phenotype based and chemical feature based BBB permeability prediction methods

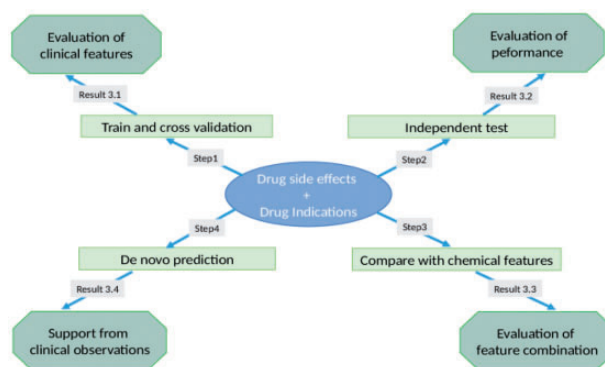


Fig. 2. Schema of experimental steps for prediction of drugs' permeability to Blood-Brain-Barrier.

2.1 Drug datasets

2.1.1 Drug dataset based on brain/blood concentration ratio

As there is no complete list of market drugs log[Brain]/[Blood] (logBB) data available, we gathered as much information as possible from five academic papers containing experimental logBB data as following: Four papers contain compounds' name without Pubchem CID (Abraham *et al.*, 2006; Li *et al.*, 2005; Subramaniana and Kitchen, 2003; Winkler and Burden, 2004). The corresponding Pubchem CID was fetched from the Pubchem auxiliary information file (ftp.ncbi.nlm.nih.gov/pubchem/Compound/Extras/CIDSynonym_filtered.gz), which lists all the names associated with each particular CID. For the paper containing Pubchem CID without compound's name (Wang *et al.*, 2015), the corresponding 'preferred' name was fetched from the same auxiliary information file which arranged the 'best' name in the first place for each CID.

Next, the above drugs were searched against the SIDER drug side effects and indications database (Kuhn *et al.*, 2015, <http://sideeffects.embl.de/>) with Pubchem CID. The drugs which exist in the SIDER database were served for clinical phenotypes based training and cross validation. This drug dataset is given in the Supplementary Table S1. *Drug Dataset Based on Brain Blood Ratio* (213 drugs in total). Based on the splitting criteria of Li and coworkers (Li *et al.*, 2005), drugs were divided into BBB+ (139 drugs) and BBB- (74 drugs) groups according to whether the brain to blood concentration ratio was greater than 0.1 or less than 0.1 respectively (i.e. $\log BB > -1$ or < -1).

2.1.2 Drug dataset based on activities in CNS

Doniger and coworkers composed a drug dataset of 179 BBB+ and 145 BBB- compounds according to their activities in CNS (Doniger *et al.*, 2002). The compounds which exist in the SIDER database (76 BBB+ and 85 BBB- (given in the Supplementary Table S2. *Drug Dataset Based on CNS Activities*) were used as independent test dataset for clinical phenotypes based prediction as well as training and cross validation through combination of drug clinical phenotypes and chemical features as this dataset contains raw data of chemical structure features.

2.2 Features extraction for drug clinical phenotypes

The drug clinical phenotypes (side effects and indications) in the SIDER database (version 4.1) are formatted according to the Medical Dictionary for Regulatory Activities (MedDRA, <http://www.meddra.org/>). The phenotypes in MedDRA are organized in a 5-level hierarchic structure: Lowest Level Term (LLT), Preferred Term (PT), High Level Term (HLT), High Level Group Terms (HLGT) and System Organ Classes (SOC). PT is a distinct descriptor for symptom, sign, diagnosis, therapeutic indication, investigation, procedure and medical social/family history characteristic.

The Preferred Terms associated with each drug were used for extracting distinct and unambiguous clinical phenotypes and for mapping to the 43 HLT under Nervous System Disorders SOC and Psychiatric Disorders SOC according to MedDRA Version 14 (in Supplementary Table S3. *High Level Group Terms for CNS Symptoms*). All the Preferred Terms under each of the 43 HLT were aggregated as corresponding set according to MedDRA Version 14. The side effects and indications (in the PT level) associated with each drug were used for matching the terms in each of the 43 HLT sets composed of Preferred Terms. Next, each drug was recorded with its number of matching times under each specific HLT group as training features.

2.3 Experiments

2.3.1 Training and cross evaluation with drug clinical phenotypes

The drug clinical phenotypes associated with the training dataset in Section 2.1.1 were used to build classification model with support vector machine (SVM) algorithm using API to Weka data mining software (version 3.6.13, Hall *et al.*, 2009). Monte Carlo method was utilized for evaluation of the performance of SVM prediction model as following: the drug dataset with side effects and/or drug indications in CNS was randomly split 1000 times into mutually exclusive training set (70%) and validation set (30%) with stratification for BBB permeability. The training dataset in each split was used for building classification model while the validation dataset was used for evaluating the model (Karalis *et al.*, 2016). The average Accuracy, AUC and F₁ score (macro weighted) in the 1000 validation sets were used for general performance evaluation of models.

We tested several common kernels (polynomial-, normalized polynomial- and radial basis function) associated with SVM for setting optimization and found the polynomial kernel having the best performance in our dataset (results shown in Supplementary Table S4. *Performance of Different Kernels in SVM Models*), hence polynomial kernel was utilized in this research project.

2.3.2 Training and cross evaluation through combination of chemical features and clinical phenotypes

The nine physico-chemistry features adopted by Doniger *et al.* (2002) are used for chemical feature based drug BBB permeability prediction. Values of the nine chemical features of the 161 drugs (in Section 2.1.2) were fetched from the same paper. The drug clinical phenotypes of the 161 market drugs were fetched from SIDER. The list of drugs with both chemical features and clinical phenotypes were performed with the same Monte Carlo random split and evaluation method as Section 2.3.1.

2.3.3 De novo drug BBB permeability prediction through clinical phenotypes

The drugs listed in the SIDER database were predicted for their BBB permeability after removing the ones in dataset 1 and dataset 2 (Section 2.1), which have prior knowledge of BBB permeability and were used for training and test. A total number of 1128 drugs were selected accordingly. The side effects and indications of the 1128 drugs were fetched as Section 2.2. Afterwards, the model trained with drug clinical phenotypes was applied to the 1128 drugs. The prediction results are shown in Table S6. *BBB Permeability Prediction for the Drugs in SIDER Database*.

2.4 Evaluation

2.4.1 Cross evaluation of performance

The cross evaluation was performed at the same time of model training (Sections 2.3.1 and 2.3.2).

2.4.2 Evaluation of performance in independent test dataset

The model trained with SVM (polynomial-kernel) from the drug clinical phenotypes of the 213 training drugs (Section 2.1.1) is applied to the independent dataset of 161 drugs (Section 2.1.2). The Accuracy, AUC and F₁ score (macro weighted) in the independent dataset were used for evaluating the prediction performance.

2.4.3 Evaluation of de novo BBB permeability prediction

The *de novo* prediction results (section 2.3.3) were analyzed through two renowned and comprehensive drug information databases

Is it possible to move "nym-" to earlier line?

(DrugBank and Drugs.com) as the following: drug was classified as BBB+ if (i) it has functions or activities in CNS; (ii) it is simple nutrient or analog of simple nutrient; (iii) it is steroid hormone. On the contrary, if drugs are categorized as vessel imaging agents or insoluble/indigestible agents with GI track administration, those drugs are BBB- by default (Davson, 1989). The statistics of the above drug category information were used to evaluate the performance of *de novo* drug BBB permeability prediction.

3 Experiments and results

3.1 Combination of drug side effects and indications is useful for BBB permeability prediction

We first built and validated the drug BBB permeability prediction model from drug side effects and drug indications in this section. A total number of 213 market drugs (drug dataset 1, in Supplementary Table S1) with known brain to blood concentration ratios were exploited (details in Section 2.1.1). These 213 drugs were divided into 139 BBB+ and 74 BBB- groups according to their logBB values. The side effects and indication of these market drugs were fetched from the SIDER database version 4.1. First, the clinical phenotypes (Preferred Term level) of the 213 market drugs in the Table S1 were mapped to the 43 subgroups under the Nervous System Disorders and Psychiatric Disorders SOC (details in Method Section 2.2). Next, the list of drugs was randomly split 1000 times into mutually exclusive training set (70%) and validation set (30%) with stratification for drug BBB permeability. The training dataset in each split was utilized for building classification model while the validation dataset was used for assessment of the model performance (Karalis et al., 2016).

The Support Vector Machine (SVM) algorithm has been extensively used in drug BBB permeability prediction (Doniger et al., 2002; Li et al., 2005; Zhang et al., 2015) and was employed in this research (with polynomial-kernel) through a widely-used data mining software Weka (version 3.6.13, Hall et al., 2009). The average of the Accuracy (A.), the area under the ROC curve (AUC) and F_1 scores (macro weighted F_1 from both positive and negative classes as well as F_1 from BBB positive drugs and F_1 from negative samples) in the 1000 randomly split was used to evaluate the general prediction performance and the results were shown in Table 1. Through such Monte Carlo cross validation method, the drug side effects achieved Accuracy 73.9%, AUC 0.709 and macro weighted F_1 score 0.737 in the drug BBB permeability prediction. To check if drug indications have similar predication power as drug side effects and to evaluate the performance with the combination of drug side effects and drug

indications, the drugs in dataset 1 with indication alone and with both side effects and indication went through the same splitting, training and cross evaluation procedure as above. The average Accuracy, AUC and F_1 scores in these experiments was also shown in Table 1. The results demonstrated that performance of drug side effects is better than drug indications (t -test, $P < e^{-11}$ for all of Accuracy, AUC and macro weighted F_1 score); while the performance of combination of side effects and indications (Accuracy = 76.0%, AUC = 0.739 and macro weighted F_1 score = 0.760) was better than either side effects (t -test, $P < e^{-5}$) or indications (t -test, $P < e^{-79}$) in the Monte Carlo cross validation datasets.

Li et al. summarized the overall accuracy in many chemical features based BBB prediction research and demonstrated the accuracy fell into the range from 71.0% to 83.7% in their cross validation datasets (reviewed by Li et al., 2005). Our clinical phenotypes based approach has comparable performance as those chemical features based methods.

3.2 Clinical phenotype based approach achieved high performance on independent dataset

We next evaluated the model trained through drug clinical phenotypes with independent test data because independent data could provide more accurate and objective evaluation of performance. As the first step, all the drugs in dataset 1 (Section 3.1) were used for building classification SVM model with sequential minimal-optimization (SMO) algorithm (Bouckaert, 2013; Hall et al., 2011; Platt, 1998) and the linear kernel was adopted as default for polynomial kernel in Weka-SMO (Bouckaert, 2013). The weight for each specific HLGT group was obtained through SMO after optimization until converge (Bouckaert, 2013; Platt, 1998). The most weighted (top 20%) HLGT groups in this SVM model are shown in Table 2 and the least weighted (bottom 20%) HLGT groups in this SVM model are shown in Supplementary Table S5.

Generally, such features meet with our expectation: Drug induced side effects of seizures, psychiatric disorders and schizophrenia were weighted high, we believe the drugs could be BBB+ if they cause such severe symptoms. It's also reasonable that the side effects of spinal cord and nerve root disorders weighted high as these diseases are organic CNS diseases. We assume the symptoms such like sleep disturbance and impulse control disorders are common side effects of BBB+ drugs. It's also commonly acknowledged that the drugs having indications for deliria-like symptoms are BBB+. On the contrary, side effects like demyelinating disorder weighted low since it could happen in both central and peripheral nervous system (Hartung et al., 1992). Drug indications for certain CNS

Table 1. Performance of side effects & indications in SVM model

T/P	Side Effects (SE)		Indications (I)		SE + I	
	Training	Validation	Training	Validation	Training	Validation
A.	.863±.030	.739±.023	.725±.033	.661±.016	.905±.026	.760±.021
AUC	.841±.038	.709±.031	.640±.095	.585±.077	.894±.031	.739±.029
$F_1^{\&}$	.861±.031	.737±.024	.675±.063	.606±.046	.905±.026	.760±.021
$F_{1(+)}^{\&}$	.895±.026	.802±.020	.800±.044	.754±.053	.926±.022	.814±.018
$F_{1(-)}^{\#}$	.796±.052	.615±.048	.421±.230	.320±.216	.863±.040	.658±.040

Each data field shows average ± std of 1000 random splits of drugs known permeability

$\&$: macro weighted F_1 from total samples;

$+$: positive samples;

$\#$: negative samples.

Table 2. The MOST weighted HLGT in the SVM model

+/-	Weight	HLGT Group Name*
	1.2917	Impulse control disorders NEC
	1.2536	Sleep disturbances (incl subtypes)
-	1.1565	Seizures (incl subtypes)
-	1.1556	I-Depressed mood disorders and disturbances
-	0.9947	I-Neuromuscular disorders
-	0.8966	Increased intracranial pressure and hydrocephalus
	0.7859	Somatoform and factitious disorders
	0.7726	I-Increased intracranial pressure and hydrocephalus
-	0.7705	I-Anxiety disorders and symptoms
-	0.7611	Psychiatric disorders NEC
-	0.7459	Mood disorders and disturbances NEC
-	0.7083	Schizophrenia and other psychotic disorders
	0.7007	Spinal cord and nerve root disorders
	0.6801	Suicidal and self-injurious behaviours NEC
	0.6528	Cranial nerve disorders (excl neoplasms)
	0.6089	I-Deliria (incl confusion)
	0.5988	Cognitive and attention disorders and disturbances

*Start with 'I-' means from drug indication, other from drug side effects
 'NEC' means 'Not Elsewhere Classified'.

diseases are also weighted low as these drugs represent only a small portion in the training drug pool and provided little differentiation power.

Doniger and coworkers curated and classified a list of drugs according to whether the drugs were active or inactive in CNS (Doniger *et al.*, 2002). This drug dataset was used as the basis of independent validation of prediction performance. Each compound in this dataset was searched against SIDER database and screened out 161 market drugs (76 BBB+ and 85 BBB- drugs, details Information in Supporting Table S2. Drug Dataset Based on CNS Activities). The drug clinical phenotypes (side effects and indication) of these 161 drugs in independent test dataset (dataset 2) were extracted from SIDER database. Then the model built from the training data was applied to this independent test dataset. The Accuracy, ROC and F₁ scores with drug clinical phenotypes in the independent dataset were shown in Table 3.

The results show that prediction with drug side effects obtained accuracy 64.6%, AUC 0.654, macro weighted F₁ score 0.639 respectively; prediction with indications obtained accuracy 54.7%, AUC 0.571 and macro weighted F₁ score 0.450 respectively; and prediction with combination of both side effect and indication achieved accuracy 68.3%, AUC 0.692 and macro weighted F₁ score 0.676 respectively. The prediction accuracy from drug clinical phenotype features is comparable to the traditional chemical features based method used in Drugbank in independent test dataset (69.5%, Cheng *et al.*, 2012; Law *et al.*, 2014; Shen *et al.*, 2010).

3.3 Combination of drug clinical phenotypes and chemical features achieved best performance

In this section we compared the performance of clinical phenotypes base approach and chemical feature based approach as well as their combination in the same dataset and found combination of these two groups of features has best performance.

The chemical features used in existing BBB permeability prediction models span wide range of geometric, connective, electrotopological and shape descriptors, however many of them depend on special database or complex software for computation and are not publically accessible (Abraham *et al.*, 2006; Li *et al.*, 2005; Norinder and Haeblerlein, 2002; Wang *et al.*, 2015). In addition,

Table 3. Prediction performance in independent dataset

T/P	Side Effects(SE)		Indications		SE + Indications	
	Training	Prediction	Training	Prediction	Training	Prediction
A.	.798	.646	.685	.547	.808	.683
AUC	.779	.654	.550	.571	.789	.692
F ₁ ^{&}	.798	.639	.592	.450	.808	.676
F ₁₍₊₎	.845	.682	.805	.676	.852	.718
F ₁₍₋₎	.711	.601	.193	.247	.725	.638

The independent prediction evaluation dataset from Doniger *et al.*

[&]: total samples macro weighted F₁;

⁺: positive samples;

⁻: negative samples.

there is no consensus among scientists on which descriptors should be used (Khanapure *et al.*, 2014; Kumar *et al.*, 2013; Li *et al.*, 2005). Here, we use the same physico-chemistry features employed by Doniger and coworkers (Doniger *et al.*, 2002), including molecular weight, volume, surface area, percent of hydrophilic surface, logP, hydrophilic-lipophilic balance, hydrogen bond donors/acceptors and 3D hydrogen bonding. Values of these features in their paper were fetched from ChemSW, which is a legacy database and is not available to public anymore. Fortunately, the values associated with each compound were published in their paper and could be utilized by us. Next, the drug clinical phenotypes of the 161 market drugs were fetched from SIDER. The list of drugs with clinical phenotypes and/or chemical features of the two groups of features were randomly split into training and validation sets 1000 times with stratification.

The prediction results were shown in Table 4. The average Accuracy, AUC and macro weighted F₁ score in the chemical features based prediction in the cross validation sets is 0.729, 0.733 and 0.725 respectively and is 0.824, 0.823 and 0.824 respectively in the clinical phenotypes based prediction. The results demonstrate that the drug clinical phenotypes features have stronger prediction power than the chemical features ($P < e^{-75}$). Furthermore, combination of drug clinical phenotypes and chemical features achieved best performance (Accuracy = 0.855, AUC = 0.854, macro weighted F₁ score = 0.854) among all tested descriptors groups. In addition, the performance by combination of both drug clinical phenotypes and chemical features in our research is better than the performance in most of the chemical features based prediction (71.0–83.7% in cross validation datasets, reviewed by Li *et al.*, 2005).

3.4 De novo prediction through drug clinical phenotype based approach

Our research in earlier sections was performed on the drugs with known BBB permeability, either logBB data or activities in central nervous system. The goal of this section is to predict drugs' BBB permeability without prior BBB permeability data. To achieve that, the model built by drug clinical phenotypes of the 213 training drugs (dataset 1, described in Section 2.1.1) was applied to the drugs in SIDER database, except for those drugs with known BBB permeability (dataset 1 and dataset 2, Section 2.1). The whole prediction results of the 1128 drugs are shown in the Supplementary Table S6. BBB Permeability Prediction for the Drugs in SIDER Database. We identified 110 drugs that can pass BBB (Supplementary Table S7) and 1018 drugs that cannot pass BBB, that means around 10% of the total drugs in the SIDER database (version 4.1) were predicted as BBB+.

Table 4. Performance of chemistry & phenotype based prediction

T/P	Chemistry(C)		Clin. phenotype(CP)		C+CP	
	Training	Prediction	Training	Prediction	Training	Prediction
A.	.757±.043	.729±.022	.956±.019	.824±.022	.973±.016	.855±.023
AUC	.756±.043	.733±.022	.955±.020	.823±.023	.972±.017	.854±.024
F ₁ ^{&c}	.754±.044	.725±.024	.956±.019	.824±.023	.973±.016	.854±.024
F ₁₍₊₎	.743±.057	.715±.036	.959±.018	.836±.022	.975±.015	.863±.022
F ₁₍₋₎ [#]	.761±.058	.737±.034	.952±.022	.810±.027	.971±.018	.844±.028
P [*]		< e ⁻⁹⁰		< e ⁻⁵		

Each data field shows average ± std from 1000 random split of Doniger *et al.*

^{&c}: total samples macro weighted F₁;

⁺: positive sample F₁;

[#]: negative samples F₁.

^{*} P-value for all of A., AUC and F₁^{&c} of this column to last column (C + CP), T-test.

Table 5. Category summary of the 110 predicted BBB+ drugs

Drug Category	Percentage
Direct CNS activities	78.2%
Nutrient (carbohydrate, vitamins, minerals)	5.5%
Steroid hormone	3.6%
Statins cardiovascular drugs	3.6%
Antibiotics	2.7%
Others	6.4%

Next, we set to analyze the predicted results. Although experimental logBB data about these drugs are not available currently, some drug databases have records on drug activities, functions, or category information which could be used for analyzing the prediction result made through clinical drug phenotypes. Therefore, it is still possible to analyze the performance of our prediction.

The DrugBank and Drugs.com are two renowned and comprehensive drug information databases. The information in the two drug database was employed as following: (i) if a drug has functions or activities found in CNS, it is classified as BBB+; (ii) if a drug is simple nutrient or analog of simple nutrient (vitamin, mineral nutrient, direct derivative of amino acid or simple sugar), it is classified as BBB+ because simple nutrients are essential for survival and activities of cells in CNS. (iii) Steroid hormones were taken as BBB positive here because most steroid hormones were confirmed as BBB+ (Pardridge and Mietus, 1979). The Supplementary Table S8 shows whether our predicted BBB+ drugs have clinical observations and indications from DrugBank and Drugs.com. Table 5 shows the summary of the 110 predicted BBB+ drugs listed in the Supplementary Table S7. On the other side, some drugs are vessel imaging agents or GI track administrated insoluble/indigestible agents, those drugs are BBB- by default (Davson, 1989). Table 6 listed these special drugs/agents for analysis and evaluation of our prediction.

The analysis results of the two lists of drugs are encouraging and our results are largely supported by clinical observations. The majority (78.2%) of our predicted BBB+ drugs has been found with direct CNS activities; 5.5% of the predicted BBB+ drugs are nutrients, 3.6% are steroid hormone. In total 87.3% of our predicted BBB+ drugs have CNS functions. In addition, the secondary CNS side effects could come from the 3.6% cardiovascular drugs (Statins). For the drugs that are non-absorbable, indigestible, or for blood vessel imaging (Table 6), 100% are correctly predicted as BBB-.

Table 6. The default BBB- drugs predicted as BBB impermeable

Drug Name	Drug Category	Reference
Ioxilan	Artery X-ray diagnostic contrast agent	Harnish <i>et al.</i> (1989)
Lactulose	Non-absorbable sugar for constipation	Chen <i>et al.</i> (2012)
Neomycin	Non-absorbable antibiotics	Taylor (2005)
Prussian blue	Insoluble colloids, detoxify heavy metal	Davson (1989)
Acarbose	Indigestible carbohydrate for diabetes	Ahr <i>et al.</i> (1989)
Rifaximin	Non-absorbable antibiotics	Andresen (2010)
Natamycin	Non-absorbable, topical & GI track drug	Levy (2007)
Sevelamer	Non-absorbable, prevent phosphate absorb. prevention drug	Plone <i>et al.</i> (2002)
Pentastarch	Inert plasma expander	Dieterich <i>et al.</i> (2003)

This study showed our drug clinical phenotypes based prediction could overcome limitations associated with traditional chemical attributes based methods. For example: (i) A commonly used chemical feature based rule 'sum of five' say that a compound generally cannot pass BBB if the sum of nitrogen (N) and oxygen (O) in the molecules is more than five (Pajouhesh and Lenz, 2005). This simple rule predicted that nitroglycerin could not pass BBB as the sum of N and O atoms is 12. However the degraded product and pharmacological effector of nitroglycerin is NO gas which is certainly capable of passing BBB (Abraham, 2006; Kumar *et al.*, 2013). Through the clinical phenotypes based approach, we correctly predicted nitroglycerin as BBB+. (ii) corticorelin is a 41-amino acid peptide and is the synthetic human corticotropin-releasing hormone (CRH), which works on the pituitary in CNS (Esposito *et al.*, 2002). Its size is out of the range of regular chemical feature based method, but our clinical phenotypes based method is still applicable and predicted it as BBB+. Esposito *et al.* experimentally demonstrated that corticorelin could break BBB and cause secondary changes (Esposito *et al.*, 2002). (iii) The third example is pentastarch which is a macromolecule without precisely defined structure; our clinical phenotype based method could work on it as no structure parameter is needed. (iv) The last example we discuss here is paclitaxel, a potent anticancer drug, shown to be pumped out from CNS although it could penetrate BBB initially (Fellner *et al.*, 2002). We correctly predicted paclitaxel as BBB- as no strong CNS side effects were recorded. It demonstrates our method accounts for passive diffusion as well as other complex and important mechanisms.

4 Discussion

The idea that drug physico-chemical properties affect BBB permeability has a history of more than 100 years (Saunders, 2014). Some researchers suggest simple rule based prediction method: for example, compounds with $P_{\text{octanol/water}}$ coefficient close to 3.4, molecular weight less than 400–600 Da and sum of nitrogen and oxygen atoms in the molecule is equal or less than 5 might have potential to penetrate BBB (Kortagere *et al.*, 2008; Pajouhesh and Lenz, 2005; Suenderhauf *et al.*, 2012). Other researchers used a variety of data mining algorithms based on diverse chemical features to predict BBB permeability (Carpenter *et al.*, 2014; Kumar *et al.*, 2013; Norinder and Haeblerlein, 2002; Subramaniana and Kitchen, 2003). In fact scientist have tried more than 1000 chemical descriptors, many of them depend on abstruse quantum chemistry calculation and are hard to get accurate data with current technology (Li *et al.*, 2005).

Besides the computational complexity, there is situation where chemical features are not available: (i) Certain drugs/biological agents have no precisely defined structures, e.g. pentastarch and polyclonal antibodies (Dieterich, 2013; Dimitrov and Marks, 2009); (ii) 3D structures of many of biological active macro-molecules are dynamic or not well-defined (Håkansson *et al.*, 1997; Kelley and Sternberg, 2009). There is also situation where chemical feature based method is not suitable: (iii) Prodrugs, the pharmacologically effective agents are their derivatives *in vivo* (Kumar *et al.*, 2013; Rautio *et al.*, 2008). Prediction based on the prodrug structure is not much meaningful. (iv) Some chemotherapy drugs/toxins were pumped out from CNS although they could infiltrate BBB initially (Fellner *et al.*, 2002). (v) Many nutrients, nutrients analogs and certain physiologically important macro-molecules pass BBB through complex biological active mechanisms (Banks, 2004; Crone, 1965), their BBB penetrating characteristics are very different from passive diffusion. For example, two glucose analog drugs, flu-deoxy-glucose and 2-deoxy-glucose are also uptaken by the glucose transporters to brain (Chen *et al.*, 2005; Pardridge, 2005) while the glucose isomer L-glucose cannot pass BBB (Pardridge, 2005). In the above cases, the accuracy of BBB permeability prediction would be low if the model was trained from agents that infiltrate BBB through passive diffusion. Scientists can neither pre-determine the mechanisms of how agents penetrate BBB, nor the suitability of models without support from exquisite *in vivo* experiments.

We conducted the research and explored from a different angle. We extended the scope of usage of information on drug side effects and drug indications to the computational BBB permeability prediction. We applied our model to the drugs in the SIDER database of drug side effects and indications in Section 3.4. Through this method, we identified more than 100 drugs with great possibility to be permeable to CNS from the drugs listed in SIDER4 database. It could be served as start point for experimental drug repositioning research for CNS disorders.

One special advantage of drug repositioning of FDA approved drugs to CNS diseases comes from the fact that most market drugs have gone through extensive clinical usage and abundant information about clinical drug phenotypes in central nervous system has been accumulated and recorded overtime. It is noteworthy to point out that most market drugs have to pass at least one bio-membrane before they could take effects (Pignatello *et al.*, 2011). For example, most oral drugs have to be absorbed through gastrointestinal track. We believe that's one of the main reasons why our predicted

BBB+ drugs percentage is about 10% (Section 3.4) while there are only less than 2% molecules able to pass BBB in the random chemical pools. Again, there is a lack of experimentally measured logBB values and the predicted results of BBB permeability remain to be experimentally confirmed in the future. Nevertheless, the clinical phenotypes based methodology serves as a good starting point for narrowing down potential drug list for drug repositioning to central nervous system disorders.

Despite the multiple advantages associated with our clinical phenotype based approach, a caveat about our methodology is that we cannot distinguish the side effects induced by the compound crossing BBB from the indirect secondary effects, e.g. the hormones production or secondary CNS effects from cardiovascular agents. Here we have some rough conjecture to be further studied in future research to overcome this limitation: By combining drug clinical phenotypic effects and drug chemical structure characteristics, we would be able to determine if the drugs produce the effects by crossing the BBB. For example, if a drug is ranked highly by models with clinical features and by models with chemical features respectively, we will have high confidence that this drug may produce CNS related phenotypes by crossing the BBB. On the other hand, if the drug is ranked highly by models with clinical features but not by models with chemical features, this drug might induce the effects indirectly through other means. In the future, we will test the hypothesis and classify drugs CNS effects into 'Direct-cross' or 'Indirect-induce' when high quality drug BBB permeability data and molecular structure parameters are available.

In summary, our work predicted brain penetration of drugs based on clinical features and showed the possibility of drugs to be repositioned to CNS disorders. Our method accounts for passive diffusion as well as a putative contribution by active transport and other complex mechanisms. This method not only applies to the FDA approved drugs but also applies to the tested/testing drugs, the failed drugs due to lack of expected efficiency and the withdrawn drugs due to toxic or others reasons, as long as clinical phenotype data of these drugs are available. We believe this idea would be of great interest to computational and experimental scientist working on pharmaceuticals for CNS disorders.

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